

**STRENGTH OF MATERIALS**

Time Allowed: 3 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script.
In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.
Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten):
2×10=20

- i) The strength of a beam depends on –
(a) bending moment (b) its weight ✓
(c) section modulus (d) shape of the section
- ii) If a coiled spring is cut into two halves, stiffness of the spring will be –
(a) double, (b) half, (c) same, (d) quadruple.
- iii) A beam is a structural element subjected to _____ load.
(a) axial, (b) transverse, (c) seismic, (d) none of these.
- iv) If a rod expands freely due to heating, it will develop –
(a) thermal stress, (b) bending stress, (c) compressive stress, (d) no stress.



- v) If diameter of a shaft is doubled the power transmitted capacity will be –
(a) same, (b) either twice or half, (c) four times, (d) eight times. ✓
- vi) Polar moment of inertia of a circular area of diameter 'D' –
(a) $\frac{\pi}{64} D^4$, (b) $\frac{\pi}{32} D^4$, (c) $\frac{\pi}{8} D^4$, (d) —
- vii) A strut is a member which carries –
(a) tensile load, (b) shear load, (c) compressive load, (d) both tensile and compressive load. ✓
- viii) The maximum deflection of a simply supported beam of length 'l' with a uniformly distributed load of 'w' per unit length is –
(a) —, (b) —, (c) —, (d) —
- ix) The shape of shear force diagram for a cantilever with a point load at the free end will be –
(a) rectangular, (b) curve, (c) square, (d) triangular.



- x) The intensity of shear stress at a section is _____ to the distance of the section from the axis of the shaft.
(a) inversely proportional, (b) directly proportional, (c) equal, (d) parallel
- xi) The relationship between Young's modulus (E), Bulk modulus (K) and Poisson's ratio (ν) is given by –
(a) $E = 3K(1 - 2\nu)$ (b) $E = 2K(1 - 2\nu)$
(c) $E = 3K(1 + 2\nu)$ (d) $E = 2K(1 - 3\nu)$
- xii) If diameter of a long column is reduced by 30%, the percentage reduction of Euler buckling load is – (a) 59, (b) 49, (c) 8, (d) 76.
- xiii) The slenderness ratio is the ratio of effective length to least _____.
(a) ultimate load (b) actual length
(c) radius of gyration (d) factor of safety
- xiv) When the helical extension spring is subjected to axial tensile force, the type of stress induced in the spring wire is –
(a) compressive stress, (b) torsional shear stress, (c) tensile stress, (d) bending stress.
- xv) In a thin cylindrical shell subjected to an internal pressure (p), the ratio of hoop stress to longitudinal stress is – (a) $1/2$, (b) 1, (c) 2, (d) 4



GROUP-B

Answer any Five (05) questions.

2. a) Illustrate stress-strain curve for ductile (M.S) material & discuss salient points.
b) The diameter of a water main is 2 m and the thickness of its wall is 20 mm. Determine the circumferential and longitudinal stresses set up in the pipe material when water is introduced in it at a pressure of 1.5 N/mm^2 . Take, Young's modulus (E) = 200 kN/mm^2 .
4+4
3. a) Deduce the formula for the elongation of a uniform cross-sectional bar under the action of its own weight.
b) Find the force required to punch a hole of 20 mm in diameter in a metal plate of 10 mm thickness. If the ultimate strength of the plate metal in shear is 300 N/mm^2 , determine the compressive stress set up in the punch.
5+3



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4.

Find the moment of inertia of the I-section shown in Fig.1 about the horizontal axis passing through the centroid of the section.

8

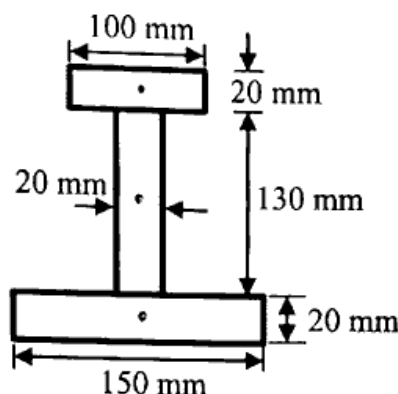


Fig.1

5.

Draw shear force and bending moment diagrams for the beam with loads as shown in the Fig.2. Show all important points in the diagrams.

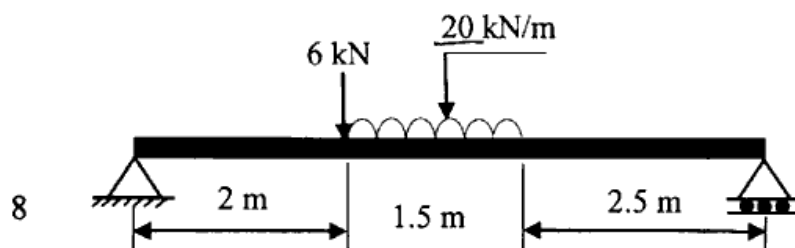


Fig. 2



6. a) What do you mean by pure bending?
 b) Deduce the bending equation: $\frac{M}{I} = \frac{\sigma}{Y} = \frac{E}{R}$ with usual notations. 2+6
7. a) A cantilever 2 m long carries a uniformly distributed load of 30 kN per metre run over its entire length. Taking, Young's modulus (E) = 2×10^4 kN/cm² and moment of inertia of the section of the cantilever about the neutral axis = 10,000 cm⁴. Determine, (i) deflection at the free end, (ii) slope at the free end.
 b) What is flexural rigidity? 6+2
8. a) What is the difference between a column and a strut?
 b) A hollow column of 200 mm external diameter and 160 mm internal diameter is used as a column of 4.5 m length. Calculate the Rankine crippling load when the column is pinned at both ends.
 Take allowable stress as 300 MPa and Rankine's constant as 1/7500. 3+5



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9. a) The diameter of a solid shaft is 250 mm. Find out the maximum safe torque which the shaft can transmit if the allowable shear stress is 4600 N/cm^2 and the maximum angle of twist is 0.25° per metre length. Take, Modulus of rigidity (G) = $8 \times 10^6 \text{ N/cm}^2$. If the shaft rotates at 200 r.p.m, what maximum power can be transmitted by the shaft? 6+2
- b) What do you mean by polar modulus?
10. a) Deduce a formula for the equivalent stiffness of helical springs arranged in series connection and parallel connection.
- b) A closed coiled helical spring having the mean coil diameter is 60mm and the diameter of the wire with which the spring is made is 5 mm. The axial load carried by the spring is 200 N. The spring is made of 20 effective turns. Taking, Modulus of rigidity (G) = 80 GPa, determine – 4+4
- (i) maximum shear stress developed in the spring, (ii) deflection of the spring.



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